

When the Label Is the Problem: Cleaning BEADs Label Noise with QLoRA Fine-Tuning on Llama-3.1-8B

Ash Dhammani, Zachary Greenman, and Abrevaa E. Prihutama

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Prof. George Harker

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Abstract. We fine-tuned Llama-3.1-8B-Instruct on the BEADs bias classification dataset using QLoRA and discovered that the model's strong reported accuracy (0.804) masks a deeper problem: the training labels are largely noise. A calibrated three-person hand-labeling exercise found that BEADs gold labels disagree with careful human judgment on 69.5% of test rows. The same model that scores 0.799 against BEADs's own labels scores only 0.283 against human consensus, which is below chance on a balanced binary task. We built a cross-dataset adapter ensemble to identify and relabel noisy training rows, retrained on the cleaned data, and recovered a stable accuracy of 0.724 against human labels (Round 2 majority), with a Round 1 peak of 0.768 at 500 training rows. This represents a lift of +44 to +49 percentage points over the noisy-data baseline. The cleaned model also transfers to external bias datasets far more effectively, reaching 0.809 on BABE. Our central finding is that the apparent silos in cross-dataset evaluation reflect labeling noise, not a fundamental disagreement about what bias means.

1. Problem and Motivation

Media bias is pervasive and consequential. Exposure to framed news content shapes political attitudes and erodes trust in institutions, yet most readers encounter that framing without any signal that it exists. Automated bias detection offers a scalable path toward surfacing that signal. Our project targets the core of that pipeline: a binary classifier labeling a sentence or headline as biased or non-biased, motivated by a media literacy tool that could operate at the point of consumption.

We chose BEADs (Bias Evaluation Across Domains) as our training data. It is the most comprehensive publicly available dataset for this task, aggregating text from eight source corpora and using a GPT-4 plus expert-review annotation pipeline (Raza et al., 2024). The dataset reports a Cohen's κ of 0.67 for its classification labels. This does not necessarily mean BEADs annotators were careless; rather, it shows that agreement under BEADs's own annotation scheme did not transfer cleanly to our public-interest definition of bias. What we did not anticipate is that the annotation quality problem is far more severe than that κ suggests. The BEADs labels are structurally misaligned with careful human judgment in ways that cause a model trained on them to learn the wrong thing entirely. Diagnosing and partially correcting that misalignment became the project's central contribution.

2. Approach

2.1 Project Timeline and Pivot

The project did not follow the path described in our original proposal. Documenting that deviation is necessary context for every methodological decision that follows.

Initial plan: Our proposal called for a standard learning curve study: fine-tune Llama-3.1-8B-Instruct on BEADs using QLoRA across five training set sizes, compare against a 3-shot prompting baseline and a TF-IDF baseline, and report how accuracy and F1 scaled with

labeled data. The goal was to establish whether a parameter-efficient fine-tuned LLM could outperform prompting on a bias classification task and at what data volume the gains stabilized.

The 3-shot baseline problem: Implementing the 3-shot baseline required selecting clean, unambiguous examples to anchor the prompt. Reviewing the BEADs data revealed the labels were not clean. A text like *"This is NOT FUNNY! This is the 2nd Amendment! YOU don't want machine gun turrets on cars, then YOU go ahead and live in your little libtard Hillary/Bernie loving threesome hiding emails..."* was labeled **non-biased**. This is not an edge case, rather it reflects a systemic pattern where BEADs's annotation criteria (focused on stereotyping and unequal treatment of demographic groups), diverges from what a careful human reader applying a public-interest definition of bias would label. We could not construct a principled 3-shot prompt because there was no coherent signal in the labels to demonstrate.

Dataset exploration: Our first instinct was to switch datasets. We reviewed BABE, cajcodes/political-bias, WNC, Article-Bias-Prediction, Hyperpartisan, and others, evaluating each for label quality, size, license, and alignment with our task definition. No single alternative was clearly superior and each had its own scope and annotation tradeoffs. This exploration reframed the problem: the issue was not unique to BEADs but reflected a broader challenge in bias annotation. Different datasets operationalize bias differently, and none offered a clean drop-in replacement.

The adapter idea: Rather than abandoning BEADs, we asked whether the cross-dataset disagreement structure we had just mapped could be turned into a cleaning signal. If independent adapters trained on BABE, cajcodes, and WNC all disagreed with a BEADs gold label, that row became a strong candidate for review. At that stage, the disagreement could have reflected either BEADs label noise or a genuine difference in how the datasets define bias, so we used blind hand-labeling to determine whether the flagged rows aligned with careful human judgment.

Defining bias and hand-labeling: Before proceeding we needed an independent ground truth to validate the ensemble. Three team members hand-labeled 500 stratified rows blind, after converging on a shared definition: a sentence is biased if it unfairly frames a public issue or event in a way that would influence a reader's perception beyond what the facts support. Personal sentiment and non-public-interest content do not qualify. We set a minimum inter-annotator agreement gate of 70% on the shared labeling subset, with a plan to recalibrate and relabel if the gate was not met. The first pass did not meet that threshold for all labeler pairs. After discussion, we identified that one labeler had been applying the bias definition too broadly. The affected labels were reviewed and corrected, after which agreement reached κ 0.57 to 0.75 and cleared the gate for proceeding. Of the 500 labeled rows, 492 were non-abstain. Those 492 rows became our human-label consensus and the independent ground truth used for all subsequent evaluation.

Pivot to cleaning: Hand-labeling confirmed the noise was severe enough to make the original learning curve study the wrong question. The project became: can label noise be identified and

corrected programmatically without hand-labeling at training scale? Ensemble-based cleaning was the path forward.

2.2 Model and Fine-Tuning Method

We fine-tuned Llama-3.1-8B-Instruct using QLoRA, quantized to 4-bit NF4 precision with LoRA adapters at rank $r=16$. Classification is framed as constrained next-token prediction: each input is formatted as an instruction-following prompt, and the model generates either *biased* or *non-biased* as the completion. Loss is masked to the label token only (completion-only loss via TRL SFTConfig). All runs used seed 42, batch size 16, and AdamW with a 0.03 warmup ratio on single H200 GPUs through Tillicum (UW Hyak HPC cluster).

2.3 Cleaning Method

The cleaning pipeline had three stages: First, we trained QLoRA adapters on three additional bias datasets (BABE, cajcodes/political-bias, and WNC) to build an independent ensemble. Second, we flagged BEADs training rows where all three non-BEADs adapters unanimously disagreed with BEADs gold. Third, we relabeled flagged rows using the ensemble's consensus vote (the flip intervention) and rebalanced to a 50/50 class split via undersampling. A second cleaning round added the Round 1 winning adapter as a fourth voter, producing strict (all four unanimous) and majority (three or more) variants.

2.4 Evaluation Measures

We evaluate against two distinct ground truths: BEADs's own noisy gold labels (the standard benchmark) and our 492-row human-label consensus. All cleaned-model results are reported against the human-label consensus. We report accuracy and macro-averaged F1. Accuracy is appropriate for the BEADs held-out test split because it is approximately balanced, with 50.1% non-biased and 49.9% biased examples. For the 492-row human-label consensus set, the class split is less perfectly controlled, so macro F1 is reported alongside accuracy to weight both classes equally regardless of support.

3. Data

The primary dataset is BEADs (shainar/BEAD, CC BY-NC 4.0). We use only the text classification bias split: 34,079 training examples and 8,520 validation examples. Because BEADs does not provide a separate test split, we carved 20% of the training set (6,816 examples) as a held-out test set before any development and used it exclusively for evaluation. The split is approximately balanced: 50.1% non-biased, 49.9% biased. For cross-dataset experiments we also used BABE (4,121 rows total), cajcodes/political-bias (Jones, 2024; 657 rows, five-class binarized to center vs. non-center), and WNC (Pryzant et al., approximately 110k rows derived from 55k bias/neutral sentence pairs, capped at 27,263 train rows to match BEADs).

4. Results

The results follow the path the project actually took: an initial learning curve that looked promising, a cross-dataset experiment that revealed a transfer problem, a hand-labeling exercise that diagnosed the cause, a cleaning intervention that corrected it, and a transfer evaluation that confirmed the fix worked.

Learning Curve

Training QLoRA adapters on nested stratified subsets of BEADs produced a clear learning curve against the noisy gold labels. The model collapses to majority-class prediction at 100 rows, establishes a meaningful decision boundary at 500, and reaches 0.804 accuracy at full training size, clearing the BEADs paper's Llama2-7B baseline of 0.77 by approximately three points. These numbers looked strong. The problem, as the subsequent experiments revealed, is that they were measuring the wrong thing.

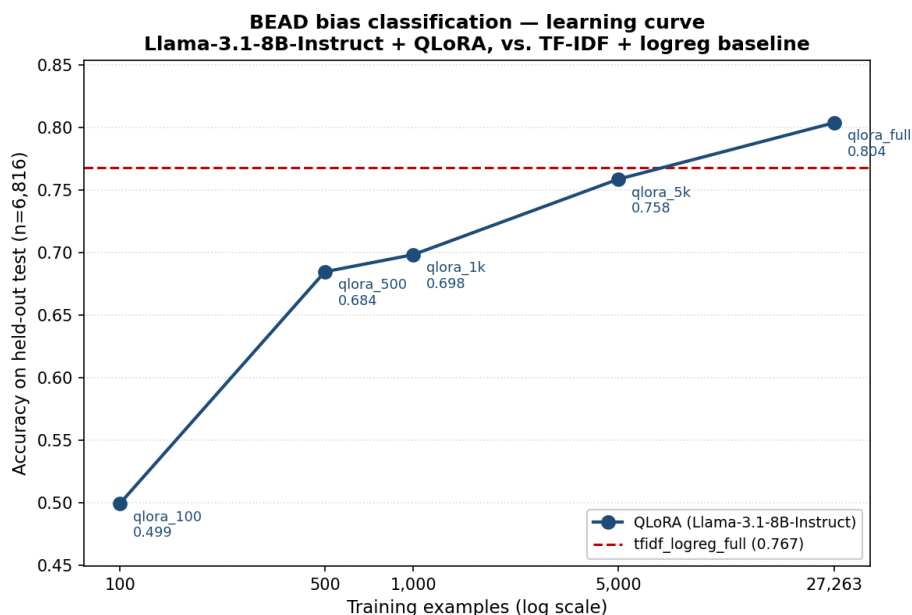


Figure 1. Accuracy on BEADs noisy gold labels by training-set size. Note: The original full-size adapter from this sweep was subsequently retrained from the same recipe and seed after a routine cleanup of intermediate outputs. The retrained adapter scored 0.799 on BEADs gold under identical settings and is the artifact used for all downstream comparisons. The 0.005 difference versus the initial 0.804 reflects bf16 training non-determinism.

Cross-Dataset Transfer

We trained QLoRA adapters on each of the four bias datasets and scored every adapter against every dataset's test split. Diagonal cells represent same-dataset performance; off-diagonal cells represent cross-dataset transfer.

Train \ Eval	BEADs	BABE	cajcodes	WNC
BEADs	0.799	0.312	0.667	0.462
BABE	0.379	0.867	0.424	0.563
cajcodes	0.422	0.588	0.970	0.557
WNC	0.407	0.760	0.576	0.734

Table 1. QLoRA cross-dataset accuracy matrix. Diagonal mean: 0.842. Off-diagonal mean: 0.510, which is statistically indistinguishable from a coin flip on a balanced binary task.

Every adapter learned its own dataset well and transferred almost nothing. The initial interpretation was that these four datasets measure four different constructs that happen to share a label vocabulary. The hand-labeling exercise that followed revealed the correct interpretation.

Label Noise Diagnosis

Three team members hand-labeled 500 stratified random rows from the BEADs test split, blind to gold labels and model predictions. Of those, 492 were non-abstain and became the human-label consensus used for all subsequent evaluation. After one recalibration pass on the public-interest topic definition of bias, the IAA gate passed.

Pair	First Pass	After Recalibration
Abrevaa-Ash	81%	81%
Abrevaa-Zach	64%	88%
Ash-Zach	63%	82%
Cohen's κ range	0.33 - 0.57	0.57 - 0.75

Table 2. Inter-annotator agreement before and after recalibration. A κ of 0.57 to 0.75 indicates moderate-to-substantial agreement.

BEADs gold labels disagree with our consensus on 69.5% of rows (342 of 492). The noise is directionally asymmetric: BEADs over-calls bias 1.3 times more often than it misses it (194 over-calls vs. 148 missed). The BEADs-fine-tuned adapter scores 0.283 against our human consensus despite scoring 0.799 against BEADs's own labels. The model did not learn to detect bias. It learned to replicate BEADs's labeling noise. This also explains the coin-flip transfer result.

Cleaning, Retraining, and Round 2

We flagged BEADs training rows where the BABE, cajcodes, and WNC adapters unanimously disagreed with BEADs gold, identifying 10,371 of 27,263 rows (38.0%) as suspect. Hand-labeling confirmed the ensemble's flip-correctness at 85.1%, clearing the gate for relabeling. We tested four cleaning variants across five training sizes, then added the Round 1 winner as a fourth voter for a second cleaning round. All results are evaluated against the 492-row human-label consensus.

Adapter	Train n	Accuracy	F1 Macro
qlora_beads_full (baseline)	27,263	0.283	0.275
Cleaned: remove + balanced	500	0.468	0.467
Cleaned: flip	27,263	0.565	0.546
Cleaned: flip + balanced (best)	500	0.768	0.768

Table 3. Key cleaning variants evaluated against the 492-row human-label consensus. The bolded row indicates the best result.

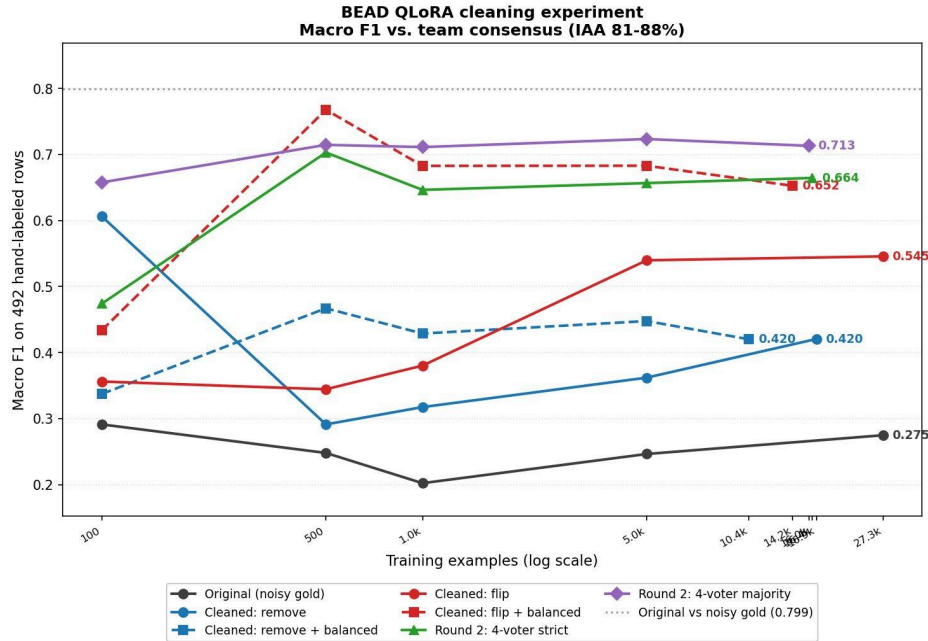


Figure 2. Macro F1 against human-label consensus across all cleaning variants and training sizes. The dotted grey line at 0.799 is the original model's score against BEADs noisy gold. The gap between that line and the black curve (original model vs. human labels, ending at 0.275) is the core finding of the project. The 4-voter majority rule (purple) produces the most stable curve across all training sizes, trading the Round 1 peak for consistency.

1. Flip beats remove: relabeling flagged rows outperforms removing them because the ensemble's directional vote reliably corrects rather than just filters.
2. Class balance matters as much as cleaning: the natural post-flip distribution is approximately 74% biased, and training on that imbalance collapses the model toward majority-class prediction; undersampling to 50/50 recovers approximately 20 percentage points.
3. Round 2 traded peak for stability: the flat majority-rule curve (approximately 0.71 to 0.72 across all sizes) is more defensible than the first round's peak-and-drop pattern, which appears to have been sampling variance rather than a stable size dependence. Evidence for this is that the majority-rule scored 0.663 at $n=100$ versus the first round's 0.492 at the same size, a gain driven entirely by cleaner labels, not more data.

Cross-Dataset Transfer After Cleaning

We evaluated all 20 cleaned BEADs adapters on the BABE, cajcodes, and WNC test splits. The results close the loop on the transfer experiment.

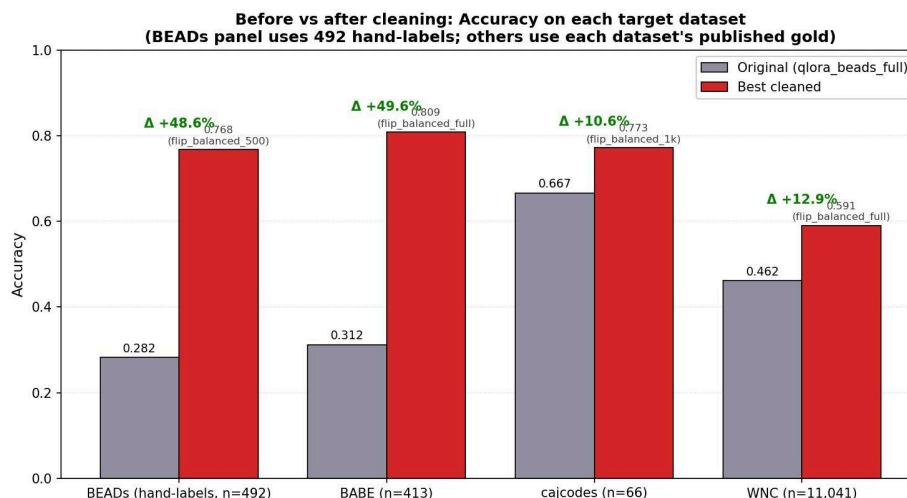


Figure 3. Accuracy before and after cleaning on each target dataset. The BEADs panel uses the 492-row human-label consensus; all other panels use each dataset's published gold labels. The cleaned adapter reaches 0.809 on BABE, within six points of a BABE-specialist model trained on BABE data directly (0.867). The poor transfer observed earlier was an artifact of label noise, not construct divergence.

Every cleaned variant beats the original adapter on every target. The BABE lift of +49.6 percentage points is the sharpest signal: original BEADs transfer to BABE was essentially chance (0.312), while cleaned BEADs transfer approaches specialist-level performance. The bias signal is more shared across datasets than the earlier result suggested.

5. Summary of Results

Model	Evaluated Against	Accuracy	F1 Macro
qlora_beads_full (noisy training)	BEADs noisy gold	0.799	0.799
qlora_beads_full (same model)	Human-label consensus	0.283	0.275
Round 1 cleaned — peak	Human-label consensus	0.768	0.768
Round 2 cleaned — stable	Human-label consensus	0.724	0.723
Round 1 cleaned — BABE transfer	BABE gold labels	0.809	0.802

Table 4. Headline accuracy figures across the project arc.

The headline lift is +48.6 percentage points from 0.283 to 0.768 against human labels. The stable post-cleaning ceiling is 0.72 to 0.77. Pushing higher would require a larger hand-labeled validation set, confidence-weighted relabeling, or active-learning-style human-in-the-loop validation.

6. Related Work and Differentiation

Bias datasets and cross-dataset transfer: BEADs (Raza et al., 2024) is our primary dataset and direct benchmark. Their fine-tuned Llama2-7B reaches 0.77 accuracy on BEADs's own labels, which our full-data adapter clears by approximately three points. Chang et al. (2024) survey cross-dataset generalization in LLM evaluation and consistently find poor transfer, attributing it to construct divergence. We replicate the poor-transfer finding but substantially revise the interpretation. Cleaning one dataset's labels against careful human judgment recovers most of the transfer gap. The silos are labeling noise, not semantic divergence.

Parameter-efficient fine-tuning: QLoRA (Dettmers et al., 2023) enables our large-scale cleaning experiment of approximately 70 adapter training runs, which would be cost-prohibitive under full fine-tuning. Prior bias classification work including BEADs's own baselines does not use this approach.

Annotation quality: Gilardi, Alizadeh, and Kubli (2023) find GPT-4 labels can match crowd-worker quality on many text classification tasks. Our findings complicate that result: on a subjective task without a strictly operationalized definition, GPT-4 labels with expert review still exhibit 69.5% disagreement with careful human judgment. Zhang et al. (2023) propose prospective active annotation to catch LLM labeling errors; we demonstrate a retrospective ensemble-based cleaning approach that recovers most accuracy lost to noise without requiring human labeling at training scale.

7. Discussion

Three aspects of this project merit reflection beyond the headline numbers.

1. The 69.5% label disagreement rate is high enough to call into question any benchmark result reported against BEADs's noisy gold alone. A model scoring 0.80 on BEADs is not necessarily a model that detects bias. It may be a model that has learned BEADs's idiosyncratic labeling decisions. This matters beyond our project: any study that fine-tunes on BEADs and evaluates on BEADs's own gold is measuring noise memorization, not bias detection.
2. The direction of the noise matters practically. BEADs over-calls bias on personal sentiment and non-public-interest content. A model trained on those labels will flag apolitical opinions as biased, which is the opposite of useful in a media literacy tool. The problem is not just that the labels are wrong. It is that they are wrong in a specific direction that undermines the downstream application.
3. The cross-dataset transfer improvement after cleaning is the project's strongest empirical signal. A single dataset cleaned against human judgment produces a model approaching specialist-level performance on an external dataset it never trained on. This confirms that the underlying construct of journalistic framing bias on public-interest topics is more coherent across datasets than the earlier transfer result suggested. Cleaning the labels made that shared signal visible.

Limitations: The hand-label set ($n=492$) limits statistical power to distinguish the two cleaning round performance ceilings. The 4.4 percentage point gap sits within the Wilson confidence interval at this sample size. The cleaning ensemble is itself trained on imperfect data, with roughly 15% of relabels wrong by human-judgment standard. This explains the non-monotonic learning curve and caps the accuracy ceiling. The public-interest topic definition of bias is defensible but contestable. A different operationalization would produce different labels and potentially different results.

8. Conclusion

We built a QLoRA fine-tuned bias classifier on Llama-3.1-8B, discovered that the training labels were substantially unreliable, cleaned them using a cross-dataset adapter ensemble, and recovered a stable +44 to +49 percentage point lift in accuracy against careful human labels. The result improved from 0.283 to a stable ceiling of 0.724 and a peak of 0.768. The cleaned model also transfers to external bias datasets far more effectively, reaching 0.809 on BABE within six points of a specialist model.

The broader implication is methodological. On any subjective NLP task annotated at scale by an LLM, reporting model accuracy against the dataset's own labels without independent ground-truth validation can substantially overstate real-world performance. Label noise is not a background nuisance. In BEADs's case it is the dominant signal, and cleaning it changed everything.

Appendix

Code, data, and model artifacts

All code, per-run evaluation metrics, figures, and the hand-labeled ground truth are available at <https://github.com/zach8421/group7-bead-qlora>. The repository contains the full training, cleaning, and evaluation pipeline; the eval_metrics.json files behind the results in Sections 4–5; the headline figures; and the 492-row human-label consensus with its blind labeling protocol in labeling/. Every result is regenerable from the committed code and frozen-split SHA-256 manifests.

Trained LoRA adapter weights are not stored in the GitHub repository because the project produced 35 adapters at approximately 97 MB each. The two headline adapters are published on Hugging Face Hub:

- Original noisy-trained baseline:
<https://huggingface.co/zachary-greenman/beads-qlora-original-noisy>
- Cleaned-data winner:
<https://huggingface.co/zachary-greenman/beads-qlora-cleaned-flip-balanced-500>

These adapters load on top of meta-llama/Llama-3.1-8B-Instruct, which requires access to the gated base model. The complete set of 35 adapters is also available on UW Tillicum at /gpfs/projects/imt526a/group7/outputs/<run_name>/adapter/ (one directory per run, e.g., /gpfs/projects/imt526a/group7/outputs/qlora_beads_full/adapter/), readable through the group7 allocation. Course staff can pull the full set with scripts/tillicum_sync.sh pull-results.

See the repository README and docs/build_log.md for a guided walkthrough of the pipeline and results.

Team Responsibilities

Abrevaa (Data, Coordination) handled data acquisition, licensing, preprocessing pipeline, data splits, and TF-IDF baseline.

Zach (Training and Modeling) handled QLoRA fine-tuning, Tillicum job scripts, hyperparameter decisions, and checkpoint management.

Ash (Evaluation and Coordination) handled F1/precision/recall, learning curve analysis, qualitative inspection, compute scheduling, and report assembly.

Why LLMs

Large language models are the natural candidate for this task. They perform well on subjective classification tasks, can be fine-tuned efficiently on domain-specific data, and handle the implicit, framing-level bias that simpler approaches miss. The main obstacle is labeled data. Bias annotation is expensive, slow, and highly subjective, making large-scale human labeling

impractical. The field has responded by using LLMs as annotators with expert review to catch errors.

Hand-Labeling Protocol

Each labeler was assigned 200 rows from the BEADs held-out test set. Labelers saw only the row `id` and `text`; they did not see BEADs's official gold label, any model predictions, or which rows were shared with other labelers. Approximately 50 rows were shared across labelers as an inter-annotator agreement check, but these rows were randomly mixed into each assignment.

Labelers filled the `label` column with exactly one of two values: `biased` or `non-biased`. Rows that could not be confidently labeled were left blank and treated as abstentions.

The working definition was: a statement is biased if it carries partisan, emotional, sarcastic, or judgmental framing on a public-interest topic. Otherwise, it is non-biased. This included sarcasm, ridicule, rhetorical questions implying a stance, loaded vocabulary on contested topics, opinion presented as fact about a contested issue, and emotional appeals on political or social topics. It did not include pure factual statements, personal logistics, off-topic content, information-seeking questions, plain opinions on uncontested topics, or neutrally quoted biased speech.

As a practical rule, labelers asked whether a careful editor would remove the sentence from a news article for being one-sided. If yes, it was labeled biased. If not, it was labeled non-biased.

After labeling, we computed agreement on the shared rows. Our pre-set gate was at least 70% agreement before proceeding. If the gate was not met, the plan was to recalibrate the definition, review disagreement patterns, and revise labels before using the consensus set for evaluation.

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